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A Practical Synthesis of Chiral Oxazolines through a Highly Diastereoselective Coupling–Cyclization Reaction of *N*-(Buta-2,3-dienyl)amides and Aryl Iodides

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Abstract: A highly *trans* diastereoselective synthesis of 2,4,5-trisubstituted oxazoline derivatives is described that proceeds through a palladium-catalyzed coupling–cyclization reaction of *N*-(buta-2,3-dienyl)amides with aromatic iodides. The reaction with optically active substrates provides convenient access to highly enantioenriched oxazoline derivatives with excellent 1,2-induction.

Keywords: amides · aryl iodides · cross-coupling · cyclization · oxazolines

Introduction

Among organic chemists, the synthesis of optically active oxazoline derivatives is of great interest because of their extensive use as chiral auxiliaries and ligands for asymmetric synthesis.[1] Moreover, numerous oxazoline-containing compounds, which include natural products and synthetic drugs, exhibit valuable pharmaceutical properties and have been widely studied in medicinal chemistry.[2] General synthetic routes to chiral oxazolines have relied on chiral amino alcohols as sources[3] and are, therefore, limited by the diversity of substituents available and the absolute configurations of the commercially available natural amino acids. Therefore, the development of efficient strategies for the preparation of multisubstituted chiral oxazolines is highly desirable. Recently, our group reported an efficient preparation of *N*-(buta-2,3-dienyl)amides from terminal *N*-propargyl amides.[4] The obtained *N*-allenyl amides were successfully converted into a variety of oxazoline derivatives via a palladium-catalyzed coupling–cyclization reaction with organic iodides.[5] As we know, a substituent α to the allene moiety always has a significant impact on the stereoselectivity of reactions that furnish five-membered cyclic compounds.[6] Consequently, we envisioned that the palladium-catalyzed coupling–cyclization reaction of α-substituted *N*-(buta-2,3-dienyl)amides may afford trisubstituted oxazoline derivatives with high diastereoselectivity through 1,2-induction. Moreover, optically active oxazolines may be easily prepared through this strategy by involving optically active *N*-(buta-2,3-dienyl)amides.

Results and Discussion

To investigate the diastereoselectivity of the coupling–cyclization reaction, *N*-(buta-2,3-dienyl)amides **4a–4f** were prepared by a Mitsunobu reaction[7] of the corresponding propargyl alcohol **1** followed by a condensation reaction with benzoyl chloride[8] and then an allenylation step with CuI/Cy_2NH[4] (Scheme 1). It is noteworthy that a Ritter reaction[9] can be used in the synthesis of (1-phenylprop-2-ynyl)-amine when R is a phenyl group.

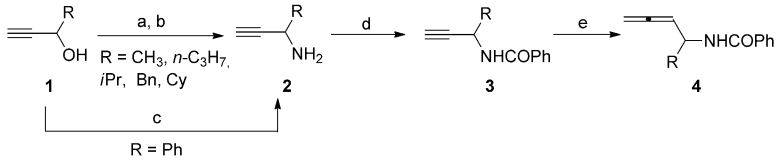
With the *N*-allenyl amides in hand, we carried out the reaction of **4** with aromatic iodides **5** under the standard conditions that included $Pd(PPh_3)_4$ (1 mol%), K_2CO_3 (2 equiv), and *N*,*N*-dimethylformamide (DMF) at 80 °C.[4] Diverse α-substituted *N*-(buta-2,3-dienyl)amides **4** and different aromatic iodides **5** were submitted to the reaction conditions (Table 1). Moderate to good yields and high diastereoselectivities were obtained in most cases. Only the reaction with **4c** afforded the corresponding oxazoline derivative **6ca** in a lower isolated yield of 67% with 20% of remaining starting material. In the instance of **4d**, the corresponding oxazoline derivative **6da** was formed with a relatively low diastereoselectivity (94:6). When R^1 is a methyl (i.e., **6aa**) or an *n*-propyl (i.e., **6ba**) group, an increase in the *trans*/*cis* ratio to 97:3 was observed. Substrates **4c**, **4e**, and **4f** performed more efficiently in the reaction to give products with very high diastereoselectivities (>98:2 *dr*). Aromatic iodides that were substituted with an acetyl (i.e., **5b**), methyl (i.e., **5c**), or bromo (i.e., **5d**) group did not alter the diastereoselectivity of the reaction. The incorporation of 2-

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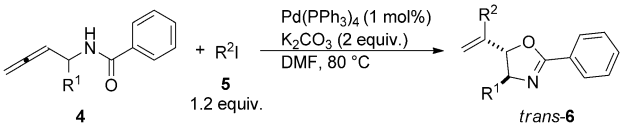
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Scheme 1. Synthesis of racemic *N*-(buta-2,3-dienyl)amides **4**. Reagents and conditions: (a) phthalimide, diethyl azodicarboxylate (DEAD), PPh_3, tetrahydrofuran (THF), 0 °C to RT; (b) $N_2H_4{\cdot}H_2O$, EtOH, reflux; (c) CH_3CN, H_2SO_4, −20 °C to RT, then HCl (1.2 N), 90 °C; (d) PhCOCl, Et_3N, CH_2Cl_2, 0 °C to RT; (e) CuI, Cy_2NH, $(HCHO)_n$, 1,4-dioxane, 100 °C.

Table 1. 1,2-Chiral induction of the coupling–cyclization reaction of substituted *N*-(buta-2,3-dienyl)amides **4** with aromatic iodides **5**.[a]



Entry	R^1 (**4**)	R^2 (**5**)	*t* [h]	Yield of *trans*-**6** [%][b]	*dr*[c]
1	CH_3 (**4a**)	Ph (**5a**)	5	89 (**6aa**)	97:3
2	*n*-C_3H_7 (**4b**)	Ph (**5a**)	6	88 (**6ba**)	97:3
3	*i*Pr (**4c**)	Ph (**5a**)	10	67 (**6ca**)	99:1[d]
4	Ph (**4d**)	Ph (**5a**)	6.5	86 (**6da**)	94:6
5	Bn (**4e**)	Ph (**5a**)	5	91 (**6ea**)	>98:2
6	Cy (**4f**)	Ph (**5a**)	14	88 (**6fa**)	98:2
7	Cy (**4f**)	4-$CH_3COC_6H_4$ (**5b**)	15	85 (**6fb**)	99:1
8	Cy (**4f**)	3-$CH_3C_6H_4$ (**5c**)	24	86 (**6fc**)	99:1[e]
9	Cy (**4f**)	4-BrC_6H_4 (**5d**)	9	82 (**6fd**)	99:1
10	Cy (**4f**)	2-thienyl (**5e**)	12	85 (**6fe**)	99:1

[a] The reaction was carried out with **4** (1.0 mmol), **5** (1.2 mmol), Pd-$(PPh_3)_4$ (1 mol%), and K_2CO_3 (2 equiv) in DMF (4 mL) at 80 °C in a Schlenk tube. [b] Isolated yield. [c] Determined by ^1H NMR analysis of the crude reaction mixture. [d] 20% of **4c** remained. [e] 5% of **4f** remained.

iodothiophene (**5e**) into the coupling–cyclization reaction also afforded product with a high level of diastereoselectivity (99:1), which is in contrast to a decline in the diastereoselectivity that was observed under Cook's conditions (Pd-$(PPh_3)_4$/K_2CO_3/tetra-*n*-butylammonium bromide (TBAB)/MeCN).[10] The relative configuration of the products were confirmed to be *trans* through a single-crystal X-ray diffraction study of compound **6da** (Figure 1).[11]

Optically active (*S*)-*N*-(buta-2,3-dienyl)amides (*S*)-**4f**, (*S*)-**4b**, and (*S*)-**4e** were then prepared and employed in the coupling–cyclization reaction to provide convenient access to the optically active oxazoline derivatives (*S*,*S*)-**6fa**, (*S*,*S*)-**6ba**, and (*S*,*S*)-**6ea** with excellent 1,2-induction and complete retention of enantiopurity (Scheme 2). The 1,2-*trans* diastereoselectivity in the cyclization step is thermodynamically controlled, in which the oxygen anion attacks from the side to ensure the *trans* orientation of the R group and vinylic unit (Scheme 3).

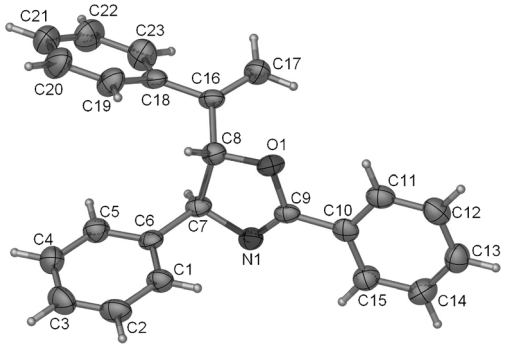
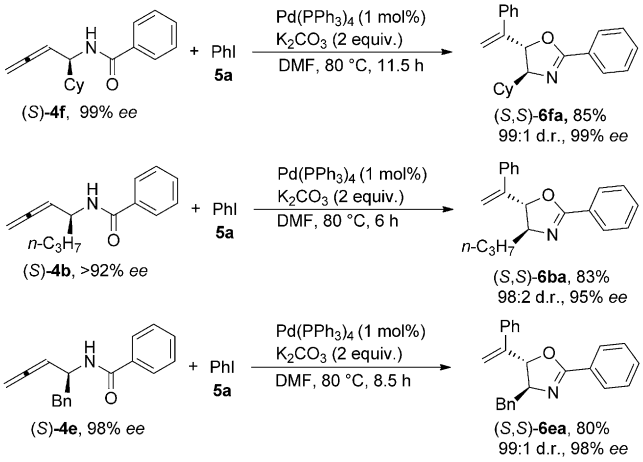
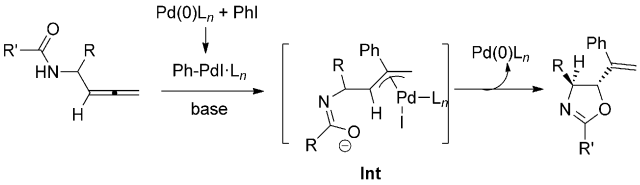


Figure 1. ORTEP representation of *trans*-**6da**.



Scheme 2. Coupling–cyclization reaction of optically active *N*-(buta-2,3-dienyl)amides **4**.



Scheme 3. A rationale to explain the diastereoselectivity.

Conclusions

To summarize, we have developed a procedure for the highly *trans* diastereoselective synthesis of trisubstituted oxazoline derivatives through the palladium-catalyzed coupling–cyclization reaction of α-substituted *N*-(buta-2,3-dienyl)amides with aromatic iodides. This method is particularly advantageous for the efficient introduction of different substituents along with excellent substrate control

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of the diastereoselectivity. Further studies in this area are currently underway in our laboratory.

Experimental Section

General Information

All reactions were carried out in oven-dried Schlenk tubes. DMF, CH_3CN, CH_2Cl_2 were dried over calcium hydride before distillation. 1,4-Dioxane and THF were dried over Na wire before distillation. All the temperatures refer to the bath temperature. The ^1H NMR chemical shifts were reported in ppm relative to TMS ($\delta = 0.00$ ppm). The ^{13}C NMR chemical shifts were reported in ppm relative to $CDCl_3$ ($\delta =$ 77.0 ppm). IR spectra were recorded on an infrared spectrometer. Melting points were measured without correction. Common reagents were purchased from commercial sources and were used without further purification. Column chromatography was performed using silica gel (300–400 mesh) and eluting with ethyl acetate and petroleum ether. TLC was performed with glass-backed silica plates. The chiral propargyl alcohols (*S*)-**4f**, (*S*)-**4b**, and (*S*)-**4e** were synthesized as reported in the literature.[12]

Synthesis of *N*-Allenyl Amides 4

*N-(Penta-3,4-dien-2-yl)benzamide (**4a**)*[4,7,8]

Typical Procedure I: To an oven-dried three-necked flask that was equipped with a dropping funnel were added phthalimide (8.87 g, 60.3 mmol), PPh_3 (15.72 g, 60.0 mmol), **1a** (3.70 mL, $d = 0.95\ g\,mL^{-1}$, 3.52 g, 50.1 mmol), and THF (100 mL). The resulting mixture was cooled to 0 °C, and DEAD (9.50 mL, $d = 1.11\ g\,mL^{-1}$, 10.51 g, 60.4 mmol) was added dropwise. After the addition, the mixture was warmed to room temperature and stirred for 27 h until the completion of the reaction as monitored by TLC. After the reaction mixture was concentrated, the residue was purified by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1) to afford 2-(but-3-yn-2-yl)-1*H*-isoindole-1,3(2*H*)-dione (6.5549 g), which was used directly in the next step without further identification.

The above prepared 2-(but-3-yn-2-yl)-1*H*-isoindole-1,3(2*H*)-dione (3.89 g, 19.5 mmol) and EtOH (60 mL) were added to a flask that was equipped with a reflux condenser. The mixture was stirred and heated at reflux. $N_2H_4{\cdot}H_2O$ (85% in H_2O, 2.50 mL, $d = 1.03\ g\,mL^{-1}$, 2.19 g, 43.7 mmol) was then added slowly. The stirring was continued for 34 min followed by filtration and evaporation to afford **2a** as the crude product. Then, crude product **2a**, CH_2Cl_2 (50 mL), and Et_3N (3.10 mL, $d = 0.73\ g\,mL^{-1}$, 2.25 g, 22.3 mmol) were added to an oven-dried three-necked flask that was equipped with a dropping funnel, and the resulting mixture was cooled to 0 °C with an ice bath. PhCOCl (2.30 mL, $d =$ $1.21\ g\,mL^{-1}$, 2.79 g, 19.8 mmol) was added dropwise at 0 °C. After the addition, the mixture was warmed to room temperature and stirred overnight. When the reaction was complete, the mixture was diluted with CH_2Cl_2 (60 mL). The resulting solution was washed with a saturated aqueous solution of $NaHCO_3$ (10 mL), HCl (1.0 N, 10 mL), and water (10 mL) and then dried with anhydrous Na_2SO_4. After filtration and evaporation, the residue was purified by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1) to afford **3a**[13] (0.68 g, 13% total yield from **1a** to **3a**, 97% purity) as a solid; m.p. 93–95 °C (*n*-hexane/diethyl ether), ref. [13a] 115 °C (benzene); ^1H NMR (300 MHz, $CDCl_3$): $\delta = 7.81$–7.75 (m, 2H, ArH), 7.56–7.40 (m, 3H, ArH), 6.29 (br s, 1H, CONH), 5.10–4.97 (m, 1H, NCH), 2.31 (d, $J = 2.1$ Hz, 1H, C≡CH), 1.53 ppm (d, $J = 7.2$ Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta =$ 166.3, 133.9, 131.6, 128.5, 127.0, 84.1, 70.6, 37.4, 22.3 ppm; IR (neat): $\tilde{\nu} =$ 3281, 1639, 1538, 1489, 1454, 1344, 1309, 1274, 1157, 1046 cm^{-1}; MS (EI): *m*/*z* (%): 173 (15.87) $[M]^+$, 105 (100).

To a flame-dried three-necked flask that was equipped with a reflux condenser were added CuI (0.31 g, 1.6 mmol), $(HCHO)_n$ (0.31 g, 10.2 mmol), **3a** (0.65 g, 3.8 mmol, 97% purity), 1,4-dioxane (20 mL), and Cy_2NH (1.50 mL, $d = 0.91\ g\,mL^{-1}$, 1.37 g, 7.5 mmol) sequentially under argon. The resulting mixture was stirred in an oil bath that was preheated to 100 °C for 12 h. After the reaction was complete as monitored by TLC, the mixture was filtered through a short pad of silica gel by eluting with ether (30 mL), and the filtrate was evaporated under reduce pressure. The resulting residue was then diluted with ether (50 mL). The solution was washed with HCl (1.0 N, 5 mL) and water (2 × 5 mL) and then dried with anhydrous Na_2SO_4. Filtration, evaporation, and purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1) afforded **4a**[10] (0.46 g, 67%) as an oil. ^1H NMR (300 MHz, $CDCl_3$): $\delta = 7.80$–7.72 (m, 2H, ArH), 7.53–7.38 (m, 3H, ArH), 6.18 (br s, 1H, CONH), 5.42–5.33 (m, 1H, C=CH), 4.96–4.88 (m, 2H, $C=CH_2$), 4.85–4.71 (m, 1H, NCH), 1.37 ppm (d, $J = 6.6$ Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta =$ 206.8, 166.6, 134.5, 131.2, 128.3, 126.8, 93.8, 78.4, 43.2, 20.5 ppm; IR (neat): $\tilde{\nu} =$ 3269, 3058, 2974, 1954, 1625, 1578, 1530, 1487, 1449, 1346, 1296 1154, 1112, 1024 cm^{-1}; MS (EI): *m*/*z* (%): 187 (9.71) $[M]^+$, 105 (100).

The following compounds were prepared according to typical procedure I with the exception of **4d**, which was synthesized as described below.

*N-(Hepta-1,2-dien-4-yl)benzamide (**4b**)*

The reaction of phthalimide (7.08 g, 48.2 mmol), PPh_3 (12.69 g, 48.4 mmol), **1b** (3.92 g, 40.0 mmol), diisopropyl azodicarboxylate (DIAD, 9.63 g, 47.7 mmol), and THF (100 mL) was carried out for 11 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1) afforded 2-(hex-1-yn-3-yl)-1*H*-isoindole-1,3(2*H*)-dione (5.63 g). The reaction of phthalimide (7.09 g, 48.2 mmol), PPh_3 (12.68 g, 48.4 mmol), **1b** (3.91 g, 39.9 mmol), DEAD (8.41 g, 48.3 mmol), and THF (100 mL) was carried out for 11 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1) afforded 2-(hex-1-yn-3-yl)-1*H*-isoindole-1,3(2*H*)-dione (8.3405 g). The two portions of 2-(hex-1-yn-3-yl)-1*H*-isoindole-1,3(2*H*)-dione were combined and used directly in the next step without further identification.

The reaction of the above prepared 2-(hex-1-yn-3-yl)-1*H*-isoindole-1,3(2*H*)-dione (13.97 g, 61.6 mmol), $N_2H_4{\cdot}H_2O$ (85% in H_2O, 7.30 mL, $d = 1.03\ g\,mL^{-1}$, 6.39 g, 127.7 mmol), and EtOH (150 mL) was carried out for 23 min to afford **2b** as a crude product. The reaction of **2b**, Et_3N (9.20 mL, $d = 0.73\ g\,mL^{-1}$, 6.68 g, 66.0 mmol), PhCOCl (7.00 mL, $d =$ $1.21\ g\,mL^{-1}$, 8.48 g, 60.3 mmol), and CH_2Cl_2 (120 mL) proceeded for 10 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1 to 5:1) afforded **3b** (2.38 g, 15% total yield from **1b** to **3b**) as an oil. ^1H NMR (300 MHz, $CDCl_3$): $\delta = 7.82$–7.75 (m, 2H, ArH), 7.53–7.37 (m, 3H, ArH), 6.53 (d, $J = 7.8$ Hz, 1H, CONH), 5.02–4.92 (m, 1H, NCH), 2.30 (d, $J = 2.4$ Hz, 1H, C≡CH), 1.83–1.69 (m, 2H, CH_2), 1.59–1.44 (m, 2H, CH_2), 0.96 ppm (t, $J = 7.4$ Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 166.4$, 133.9, 131.6, 128.5, 127.0, 83.2, 71.2, 41.5, 37.8, 18.9, 13.5 ppm; IR (neat): $\tilde{\nu} = 3291$, 1634, 1523, 1336, 1305, 1281, 1154, 1031 cm^{-1}; MS (ESI): *m*/*z*: 224 $[M+Na]^+$; HRMS: *m*/*z*: calcd for $C_{13}H_{15}NOH^+$: 202.1226 $[M+H]^+$; found: 202.1233.

The reaction of CuI (1.15 g, 6.0 mmol), $(HCHO)_n$ (0.91 g, 30.4 mmol), **3b** (2.38 g, 11.9 mmol), Cy_2NH (4.30 mL, $d = 0.91\ g\,mL^{-1}$, 3.92 g, 21.6 mmol), and 1,4-dioxane (30 mL) was carried out for 7 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1) afforded **4b** (1.38 g, 55%) as an oil. ^1H NMR (300 MHz, $CDCl_3$): $\delta = 7.81$–7.76 (m, 2H, ArH), 7.54–7.39 (m, 3H, ArH), 6.13 (br s, 1H, CONH), 5.38–5.29 (m, 1H, C=CH), 4.94–4.86 (m, 2H, $C=CH_2$), 4.78–4.65 (m, 1H, NCH), 1.77–1.55 (m, 2H, CH_2), 1.52–1.36 (m, 2H, CH_2), 0.95 ppm (t, $J = 7.2$ Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 206.9$, 166.7, 134.5, 131.1, 128.2, 126.8, 92.6, 77.9, 47.3, 37.1, 18.9, 13.7 ppm; IR (neat): $\tilde{\nu} = 3302$, 2958, 1955, 1625, 1577, 1532, 1489, 1302, 1279, 1149, 1116, 1028 cm^{-1}; MS (EI): *m*/*z* (%): 215 (1.11) $[M]^+$, 105 (100); HRMS: *m*/*z*: calcd for $C_{14}H_{17}NO$: 215.1310 $[M]^+$; found: 215.1313.

*N-(2-Methylhexa-4,5-dien-3-yl)benzamide (**4c**)*

The reaction of phthalimide (8.83 g, 60.1 mmol), PPh_3 (15.70 g, 59.9 mmol), **1c** (4.92 g, 50.2 mmol), DEAD (9.50 mL, $d = 1.11\ g\,mL^{-1}$, 10.51 g, 60.3 mmol), and THF (100 mL) was carried out overnight. Pu-

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rification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1) afforded 2-(4-methylpent-1-yn-3-yl)-1*H*-isoindole-1,3(2*H*)-dione (6.20 g), which was used directly in the next step without further identification.

The reaction of the above prepared 2-(4-methylpent-1-yn-3-yl)-1*H*-isoindole-1,3(2*H*)-dione (4.42 g, 19.5 mmol), $N_2H_4{\cdot}H_2O$ (85% in H_2O, 2.50 mL, $d = 1.03\ g\,mL^{-1}$, 2.19 g, 43.7 mmol), and EtOH (60 mL) was carried out for 27 min to afford **2c** as a crude product. The reaction of **2c**, Et_3N (3.10 mL, $d = 0.73\ g\,mL^{-1}$, 2.25 g, 22.2 mmol), PhCOCl (2.30 mL, $d = 1.21\ g\,mL^{-1}$, 2.79 g, 19.8 mmol), and CH_2Cl_2 (50 mL) was carried out overnight. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 5:1) afforded **3c** (1.14 g, 16% total yield from **1c** to **3c**) as a solid; m.p. 83–84 °C (*n*-hexane/ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 7.82$–7.75 (m, 2H, ArH), 7.56–7.41 (m, 3H, ArH), 6.27 (d, $J = 7.5$ Hz, 1H, CONH), 4.93–4.85 (m, 1H, NCH), 2.30 (d, $J = 2.1$ Hz, 1H, C≡CH), 2.14–2.01 (m, 1H, CH), 1.07 (d, $J = 6.6$ Hz, 3H, CH_3), 1.06 ppm (d, $J = 6.6$ Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 166.5$, 134.1, 131.6, 128.6, 127.0, 81.6, 72.2, 47.5, 32.7, 18.8, 17.6 ppm; IR (neat): $\tilde{\nu} = 3315$, 3266, 2967, 1636, 1521, 1346, 1490, 1348, 1309, 1282, 1153, 1078 cm^{-1}; MS (EI): *m*/*z* (%): 201 (0.17) $[M]^+$, 105 (100). Elemental analysis calcd (%) for $C_{13}H_{15}NO$: C 77.58, H 7.51, N 6.96; found: C 77.43, H 7.50, N 6.96.

The reaction of CuI (0.48 g, 2.5 mmol), $(HCHO)_n$ (0.3795 g, 12.6 mmol), **3c** (1.01 g, 5.0 mmol), Cy_2NH (1.80 mL, $d = 0.91\ g\,mL^{-1}$, 1.64 g, 9.1 mmol), and 1,4-dioxane (25 mL) was carried out for 11 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1) afforded **4c**[10] (0.41 g, 38%) as a solid; m.p. 81–82 °C (*n*-hexane/ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 7.81$–7.75 (m, 2H, ArH), 7.55–7.40 (m, 3H, ArH), 6.16 (d, $J = 8.4$ Hz, 1H, CONH), 5.35–5.27 (m, 1H, C=CH), 4.91 (dd, $J = 6.6$, 3.6 Hz, 2H, $C{=}CH_2$), 4.67–4.57 (m, 1H, NCH), 2.09–1.94 (m, 1H, CH), 1.00 (d, $J = 6.6$ Hz, 3H, CH_3), 0.99 ppm (d, $J = 6.6$ Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 207.3$, 166.8, 134.9, 131.3, 128.5, 126.8, 90.8, 78.2, 52.3, 32.6, 18.4, 18.3 ppm; IR (neat): $\tilde{\nu} = 3292$, 2957, 1954, 1633, 1524, 1324, 1135, 1046 cm^{-1}; MS (EI): *m*/*z* (%): 215 (0.81) $[M]^+$, 105 (100).

*N-(1-Phenylbuta-2,3-dien-1-yl)benzamide (**4d**)*[4,8,9]

To a flame-dried three-necked flask that was equipped with a dropping funnel were sequentially added Na_2SO_4 (7.11 g, 50.0 mmol), MeCN (150 mL), and **1d** (6.61 g, 50.1 mmol). The resulting mixture was cooled to −20 °C, and a solution of H_2SO_4 (97%, 13.00 mL, $d = 1.84\ g\,mL^{-1}$, 23.2 g, 236.8 mmol) in MeCN (100 mL) was added dropwise over 1.5 h. After the addition, the mixture was slowly warmed to room temperature and stirred for 2 h. Then, the solution was evaporated under reduce pressure, and the concentrated solution was poured into ice water. The resulting solution was extracted with Et_2O and dried with anhydrous Na_2SO_4. Filtration, evaporation, and purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 5:1) afforded *N*-(1-phenylprop-2-yn-1-yl)acetamide (9.75 g), which was used directly in the next step without further identification.

The above prepared *N*-(1-phenylprop-2-yn-1-yl)acetamide (9.75 g, 56.3 mmol) and HCl (1.2 N, 180 mL) were added to a flask that was equipped with a reflux condenser. The resulting mixture was then stirred in an oil bath that was preheated to 90 °C for 19 h. After the reaction was complete as monitored by TLC, the resulting solution was extracted with Et_2O. The aqueous layer was adjusted to approximately pH 8 with solid $NaHCO_3$, and the resulting solution was extracted again with Et_2O. The combined organic layers were dried with anhydrous Na_2SO_4. Filtration, evaporation, and purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 1:1) afforded **2d** (3.21 g), which was used directly in the next step without further identification.

This procedure was carried out in accordance with Typical Procedure I. The reaction of **2d** (3.21 g, 24.4 mmol), Et_3N (8.70 mL, $d = 0.73\ g\,mL^{-1}$, 6.32 g, 62.4 mmol), PhCOCl (5.80 mL, $d = 1.21\ g\,mL^{-1}$, 7.02 g, 50.0 mmol), and CH_2Cl_2 (70 mL) was carried out overnight. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 20:1 to 5:1) afforded **3d**[14] (5.61 g, 48% total yield from **1d** to **3d**) as a solid; m.p. 130–131 °C (*n*-hexane/ethyl acetate), ref. [14] 84.0–85.6 °C; ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 7.82$–7.76 (m, 2H, ArH), 7.62–7.55 (m, 2H, ArH), 7.55–7.47 (m, 1H, ArH), 7.47–7.29 (m, 5H, ArH), 6.64 (d, $J = 6.9$ Hz, 1H, CONH), 6.24 (dd, $J = 8.4$, 2.4 Hz, 1H, NCH), 2.54 ppm (d, $J = 2.4$ Hz, 1H, C≡CH); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 166.2$, 138.2, 133.6, 131.8, 128.7, 128.5, 128.2, 127.1, 127.0, 81.6, 73.3, 44.9 ppm; IR (neat): $\tilde{\nu} = 3326$, 3277, 2987, 2904, 1635, 1511, 1487, 1322, 1049 cm^{-1}; MS (EI): *m*/*z* (%): 235 (53.29) $[M]^+$, 105 (100).

The reaction of CuI (1.07 g, 5.6 mmol), $(HCHO)_n$ (0.83 g, 27.6 mmol), **3d** (2.59 g, 11.0 mmol), Cy_2NH (4.00 mL, $d = 0.91\ g\,mL^{-1}$, 3.65 g, 20.1 mmol), and 1,4-dioxane (40 mL) was carried out for 8 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1 to 5:1) afforded **4d** (1.78 g, 65%) as a solid; m.p. 91–92 °C (*n*-hexane/ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 7.81$–7.75 (m, 2H, ArH), 7.53–7.25 (m, 8H, ArH), 6.56 (d, $J = 7.2$ Hz, 1H, CONH), 5.88–5.79 (m, 1H, NCH), 5.58–5.50 (m, 1H, C=CH), 5.01 ppm (dd, $J = 6.6$, 3.6 Hz, 2H, $C{=}CH_2$); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 206.9$, 166.3, 140.9, 134.3, 131.5, 128.6, 128.5, 127.7, 127.1, 126.9, 92.9, 79.3, 51.2 ppm; IR (neat): $\tilde{\nu} = 3345$, 2988, 1953, 1631, 1517, 1486, 1314, 1080, 1031 cm^{-1}; MS (EI): *m*/*z* (%): 249 (3.95) $[M]^+$, 105 (100). Elemental analysis calcd (%) for $C_{17}H_{15}NO$: C 81.90, H 6.06, N 5.62; found: C 81.85, H 6.06, N 5.61.

*N-(1-Phenylpenta-3,4-dien-2-yl)benzamide (**4e**)*

The reaction of phthalimide (7.06 g, 48.0 mmol), PPh_3 (12.61 g, 48.1 mmol), **1e** (5.69 g, 38.9 mmol), DIAD (9.71 g, 48.1 mmol), and THF (100 mL) was carried out for 10 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1) afforded 2-(1-phenylbut-3-yn-2-yl)-1*H*-isoindole-1,3(2*H*)-dione (5.84 g). The reaction of phthalimide (7.15 g, 48.7 mmol), PPh_3 (12.60 g, 48.1 mmol), **1e** (5.85 g, 40.1 mmol), DIAD (9.76 g, 48.3 mmol), and THF (100 mL) was carried out for 17 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1) afforded 2-(1-phenylbut-3-yn-2-yl)-1*H*-isoindole-1,3(2*H*)-dione (8.53 g). The two portions of 2-(1-phenylbut-3-yn-2-yl)-1*H*-isoindole-1,3(2*H*)-dione were combined and used directly in the next step without further identification.

The reaction of the above prepared 2-(1-phenylbut-3-yn-2-yl)-1*H*-isoindole-1,3(2*H*)-dione (12.93 g, 45.0 mmol), $N_2H_4{\cdot}H_2O$ (85% in H_2O, 5.70 mL, $d = 1.03\ g\,mL^{-1}$, 4.99 g, 99.7 mmol), and EtOH (130 mL) was carried out for 20 min to afford **2e** as a crude product. The reaction of **2e**, Et_3N (7.20 mL, $d = 0.73\ g\,mL^{-1}$, 5.26 g, 51.9 mmol), PhCOCl (5.50 mL, $d = 1.21\ g\,mL^{-1}$, 6.66 g, 47.4 mmol), and CH_2Cl_2 (100 mL) was carried out for 5 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 5:1) afforded **3e** (5.57 g, 31% total yield from **1e** to **3e**) as a solid; m.p. 103–105 °C (*n*-hexane/ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 7.72$ (d, $J = 7.2$ Hz, 2H, ArH), 7.54–7.22 (m, 8H, ArH), 6.33 (br s, 1H, CONH), 5.29–5.19 (m, 1H, NCH), 3.13 (dd, $J = 13.2$, 5.1 Hz, 1H, one proton in CH_2Ph), 3.06 (dd, $J = 13.2$, 6.9 Hz, 1H, one proton in CH_2Ph), 2.32 ppm (d, $J = 2.4$ Hz, 1H, C≡CH); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 166.3$, 136.1, 133.8, 131.7, 129.9, 128.6, 128.3, 127.0, 126.9, 82.3, 72.7, 42.7, 41.0 ppm; IR (neat): $\tilde{\nu} = 2987$, 1634, 1528, 1339, 1307, 1263 1065 cm^{-1}; MS (EI): *m*/*z* (%): 249 (2.64) $[M]^+$, 105 (100). Elemental analysis calcd (%) for $C_{17}H_{15}NO$: C 81.90, H 6.06, N 5.62; found: C 81.66, H 6.04, N 5.54.

The reaction of CuI (1.76 g, 9.3 mmol), $(HCHO)_n$ (1.37 g, 45.6 mmol), **3e** (4.45 g, 17.9 mmol), Cy_2NH (6.40 mL, $d = 0.91\ g\,mL^{-1}$, 5.84 g, 32.2 mmol), and 1,4-dioxane (45 mL) was carried out for 6 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1) afforded **4e**[10] (3.05 g, 64%) as a solid; m.p. 87–88 °C (*n*-hexane/ethyl acetate); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 7.70$ (d, $J = 7.2$ Hz, 2H, ArH), 7.53–7.36 (m, 3H, ArH), 7.36–7.19 (m, 5H, ArH), 6.17 (d, $J = 6.9$ Hz, 1H, CONH), 5.38–5.29 (m, 1H, C=CH), 5.04–4.80 (m, 3H, $C{=}CH_2$ and NCH), 3.07 (dd, $J = 13.8$, 5.6 Hz, 1H, one proton of CH_2Ph), 2.99 ppm (dd, $J = 13.5$, 7.2 Hz, 1H, one proton of CH_2Ph); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 206.8$, 166.6, 137.1, 134.5, 131.4, 129.6, 128.5, 128.3, 126.8, 126.5, 91.9, 78.7, 48.3, 40.9 ppm; IR (neat): $\tilde{\nu} = 3306$, 1954, 1636, 1529, 1352, 1282, 1200, 1083 cm^{-1}; MS (EI): *m*/*z* (%): 263 (0.49) $[M]^+$, 105 (100).

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*N-(1-Cyclohexylbuta-2,3-dien-1-yl)benzamide (**4f**)*

The reaction of phthalimide (12.50 g, 85.0 mmol), PPh_3 (21.12 g, 80.6 mmol), **1f** (9.41 g, 68.2 mmol), DEAD (14.60 g, 83.9 mmol), and THF (175 mL) was carried out overnight. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 20:1) afforded 2-(1-cyclohexylprop-2-ynyl)-1*H*-isoindole-1,3(2*H*)-dione (6.44 g), which was used directly in the next step without further identification.

The reaction of the above prepared 2-(1-cyclohexylprop-2-ynyl)-1*H*-isoindole-1,3(2*H*)-dione (6.44 g, 24.1 mmol), $N_2H_4{\cdot}H_2O$ (85% in H_2O, 3.00 mL, $d = 1.03\ g\,mL^{-1}$, 2.63 g, 52.5 mmol), and EtOH (65 mL) was carried out for 20 min to afford **2f** as a crude product. The reaction of **2f**, Et_3N (3.70 mL, $d = 0.73\ g\,mL^{-1}$, 2.69 g, 26.5 mmol), PhCOCl (2.80 mL, $d = 1.21\ g\,mL^{-1}$, 3.39 g, 24.1 mmol), and CH_2Cl_2 (60 mL) was carried out for 6 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1) afforded **3f** (2.73 g, 16% total yield from **1f** to **3f**) as a solid; m.p. 143–143 °C (*n*-hexane/ethyl acetate); ^1H NMR (300 MHz, $CDCl_3$): $\delta = 7.82$–7.75 (m, 2H, ArH), 7.55–7.39 (m, 3H, ArH), 6.35 (d, $J = 8.1$ Hz, 1H, CONH), 4.92–4.83 (m, 1H, NCH), 2.30 (d, $J = 2.4$ Hz, 1H, C≡CH), 1.95–1.60 (m, 6H, Cy), 1.35–1.07 ppm (m, 5H, Cy); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 166.4$, 134.0, 131.6, 128.5, 127.0, 82.0, 72.1, 46.8, 42.2, 29.2, 28.3, 26.1, 25.8, 25.7 ppm; IR (neat): $\tilde{\nu} =$ 3320, 3268, 2929, 2852, 1633, 1522, 1489, 1453, 1339, 1265 cm^{-1}; MS (EI): *m*/*z* (%): 241 (0.83) $[M]^+$, 105 (100). Elemental analysis calcd (%) for $C_{16}H_{19}NO$: C 79.63, H 7.94, N 5.80; found: C 79.55, H 7.93, N 5.78.

The reaction of CuI (0.96 g, 5.1 mmol), $(HCHO)_n$ (0.76 g, 25.3 mmol), **3f** (2.42 g, 10.0 mmol), Cy_2NH (3.60 mL, $d = 0.91\ g\,mL^{-1}$, 3.28 g, 18.1 mmol), and 1,4-dioxane (25 mL) was carried out for 8 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 20:1 to 10:1) afforded **4f** (1.53 g, 60%) as a solid; m.p. 140–141 °C (*n*-hexane/ethyl acetate); ^1H NMR (300 MHz, $CDCl_3$): $\delta = 7.81$–7.74 (m, 2H, ArH), 7.54–7.39 (m, 3H, ArH), 6.17 (d, $J = 8.7$ Hz, 1H, CONH), 5.36–5.27 (m, 1H, C=CH), 4.90 (dd, $J = 6.6$, 3.6 Hz, 2H, C=CH_2), 4.66–4.55 (m, 1H, NCH), 1.88–1.57 (m, 6H, Cy), 1.33–0.97 ppm (m, 5H, Cy); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 207.1$, 166.7, 134.9, 131.3, 128.5, 126.8, 91.1, 78.1, 51.8, 42.6, 29.1, 28.8, 26.3, 26.1, 26.0 ppm; IR (neat): $\tilde{\nu} = 3277$, 2917, 2849, 1954, 1632, 1528, 1336, 1043 cm^{-1}; MS (EI): *m*/*z* (%): 255 (2.25) $[M]^+$, 105 (100). Elemental analysis calcd (%) for $C_{17}H_{21}NO$: C 79.96, H 8.29, N 5.49; found: C 79.95, H 8.32, N 5.48.

*N-[(1S)-1-Cyclohexylbuta-2,3-dien-1-yl]benzamide ((S)-**4f**)*

The reaction of phthalimide (3.89 g, 26.4 mmol), PPh_3 (6.91 g, 26.4 mmol), (*R*)-**1f** (3.00 g, 21.7 mmol, 99% *ee*), DEAD (4.60 g, 26.4 mmol), and THF (60 mL) was carried out for 16 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 20:1) afforded 2-[(1*S*)-1-cyclohexylprop-2-ynyl]-1*H*-isoindole-1,3(2*H*)-dione (3.64 g), which was used directly in the next step without further identification.

The reaction of the above prepared 2-[(1*S*)-1-cyclohexylprop-2-ynyl]-1*H*-isoindole-1,3(2*H*)-dione (2.97 g, 11.1 mmol), $N_2H_4{\cdot}H_2O$ (85% in H_2O, 1.50 mL, $d = 1.03\ g\,mL^{-1}$, 1.31 g, 26.2 mmol), and EtOH (35 mL) was carried out for 25 min to afford (*S*)-**2f** as a crude product. The reaction of (*S*)-**2f**, Et_3N (2.00 mL, $d = 0.73\ g\,mL^{-1}$, 1.45 g, 14.3 mmol), PhCOCl (1.40 mL, $d = 1.21\ g\,mL^{-1}$, 1.70 g, 12.1 mmol), and CH_2Cl_2 (30 mL) was carried out for 10 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 30:1) afforded (*S*)-**3f** (0.85 g), which was submitted to the next step without further identification.

The reaction of CuI (0.26 g, 1.3 mmol), $(HCHO)_n$ (0.20 g, 6.8 mmol), (*S*)-**3f** (0.65 g, 2.7 mmol), Cy_2NH (1.00 mL, $d = 0.91\ g\,mL^{-1}$, 0.91 g, 5.0 mmol), and 1,4-dioxane (5 mL) was carried out for 8.5 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 20:1 to 10:1) afforded (*S*)-**4f** (0.49 g, 14% total yield from (*R*)-**1f** to (*S*)-**4f**) as a solid; m.p. 141–143 °C (*n*-hexane/ethyl acetate); HPLC analysis (Chiralcel IC column, hexane/*i*PrOH = 90:10, 1.0 mL min^{-1}, $\lambda =$ 214 nm): $t_R = 10.66$ min (major) and $t_R = 12.91$ min (minor), 99% *ee*; $[\alpha]_D^{20} = +52.9$ ($c = 0.98$, $CHCl_3$); ^1H NMR (400 MHz, $CDCl_3$): $\delta = 7.78$ (d, $J = 7.2$ Hz, 2H, ArH), 7.52–7.39 (m, 3H, ArH), 6.25 (d, $J = 8.8$ Hz, 1H, CONH), 5.34–5.27 (m, 1H, C=CH), 4.89 (dd, $J = 6.4$, 3.2 Hz, 2H, C=CH_2), 4.63–4.55 (m, 1H, NCH), 1.87–1.71 (m, 4H, Cy), 1.71–1.58 (m, 2H, Cy), 1.32–1.00 ppm (m, 5H, Cy); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta =$ 207.1, 166.7, 134.8, 131.3, 128.5, 126.8, 91.1, 78.1, 51.9, 42.6, 29.1, 28.8, 26.3, 26.1, 26.0 ppm; IR (neat): $\tilde{\nu} = 3278$, 2931, 2848, 1953, 1632, 1526, 1336, 1316 cm^{-1}; MS (EI): *m*/*z* (%): 255 (1.95) $[M]^+$, 105 (100). Elemental analysis calcd (%) for $C_{17}H_{21}NO$: C 79.96, H 8.29, N 5.49; found: C 79.94, H 8.14, N 5.16.

*N-[(4S)-Hepta-1,2-dien-4-yl]benzamide ((S)-**4b**)*

The reaction of phthalimide (6.54 g, 44.5 mmol), PPh_3 (11.63 g, 44.3 mmol), (*R*)-**1b** (3.72 g, 37.9 mmol, 93% *ee*), DEAD (7.74 g, 44.5 mmol), and THF (100 mL) was carried out overnight. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 15:1) afforded 2-[(3*S*)-hex-1-yn-3-yl]-1*H*-isoindole-1,3(2*H*)-dione (4.73 g), which was used directly in the next step without further identification.

The reaction of the above prepared 2-[(3*S*)-hex-1-yn-3-yl]-1*H*-isoindole-1,3(2*H*)-dione (4.23 g, 18.6 mmol), $N_2H_4{\cdot}H_2O$ (85% in H_2O, 2.20 mL, $d = 1.03\ g\,mL^{-1}$, 1.93 g, 38.5 mmol), and EtOH (50 mL) was carried out for 13 min to afforded (*S*)-**2b** as a crude product. The reaction of (*S*)-**2b**, Et_3N (3.00 mL, $d = 0.73\ g\,mL^{-1}$, 2.18 g, 21.5 mmol), PhCOCl (2.10 mL, $d = 1.21\ g\,mL^{-1}$, 2.54 g, 18.1 mmol), and CH_2Cl_2 (45 mL) was carried out overnight. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1 to 5:1) afforded (*S*)-**3b** (0.48 g), which was submitted to the next step without further identification.

The reaction of CuI (0.19 g, 1.0 mmol), $(HCHO)_n$ (0.15 g, 5.0 mmol), (*S*)-**3b** (0.41 g, 2.0 mmol), Cy_2NH (0.70 mL, $d = 0.91\ g\,mL^{-1}$, 0.64 g, 3.5 mmol), and 1,4-dioxane (3 mL) was carried out for 15.5 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1) afforded (*S*)-**4b** (0.19 g, 3% total yield from (*R*)-**1b** to (*S*)-**4b**) as an oil. HPLC analysis (Chiralcel IC column, hexane/*i*PrOH = 90:10, 1.0 mL min^{-1}, $\lambda = 214$ nm): $t_R = 11.05$ min (major) and $t_R =$ 11.85 min (minor), >92% *ee*; $[\alpha]_D^{20} = -32.9$ ($c = 0.98$, $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): $\delta = 7.81$–7.74 (m, 2H, ArH), 7.53–7.37 (m, 3H, ArH), 6.19 (d, $J = 7.8$ Hz, 1H, CONH), 5.37–5.29 (m, 1H, C=CH), 4.94–4.85 (m, 2H, C=CH_2), 4.78–4.64 (m, 1H, NCH), 1.77–1.55 (m, 2H, CH_2), 1.52–1.36 (m, 2H, CH_2), 0.95 ppm (t, $J = 7.4$ Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 207.0$, 166.7, 134.7, 131.3, 128.5, 126.8, 92.8, 78.3, 47.2, 37.3, 19.0, 13.8 ppm; IR (neat): $\tilde{\nu} = 3291$, 2954, 1959, 1637, 1537, 1319, 1282, 1167 cm^{-1}; MS (ESI): *m*/*z*: 216 $[M+H]^+$; HRMS: *m*/*z*: calcd for $C_{14}H_{17}NOH^+$: 216.1383 $[M+H]^+$; found: 216.1384.

*N-[(2S)-1-Phenylpenta-3,4-dien-2-yl]benzamide ((S)-**4e**)*

The reaction of phthalimide (1.33 g, 9.0 mmol), PPh_3 (2.36 g, 9.0 mmol), (*R*)-**1e** (1.01 g, 6.9 mmol, 97% *ee*), DEAD (1.40 mL, $d = 1.11\ g\,mL^{-1}$, 1.55 g, 8.9 mmol), and THF (17 mL) was carried out for 10 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1) afforded 2-[(1*S*)-1-(phenylmethyl)-2-propyn-1-yl]-1*H*-isoindole-1,3(2*H*)-dione (1.35 g), which was used directly in the next step without further identification.

The reaction of the above prepared 2-[(1*S*)-1-(phenylmethyl)-2-propyn-1-yl]-1*H*-isoindole-1,3(2*H*)-dione (1.33 g, 4.8 mmol), $N_2H_4{\cdot}H_2O$ (85% in H_2O, 0.60 mL, $d = 1.03\ g\,mL^{-1}$, 0.53 g, 10.5 mmol), and EtOH (15 mL) was carried out for 20 min to afford (*S*)-**2e** as a crude product. The reaction of (*S*)-**2e**, Et_3N (0.90 mL, $d = 0.73\ g\,mL^{-1}$, 0.65 g, 6.5 mmol), PhCOCl (0.60 mL, $d = 1.21\ g\,mL^{-1}$, 0.73 g, 5.2 mmol), and CH_2Cl_2 (13 mL) was carried out for 9 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1 to 5:1) afforded (*S*)-**3e** (0.70 g), which was submitted to the next step without further identification.

The reaction of CuI (0.24 g, 1.3 mmol), $(HCHO)_n$ (0.19 g, 6.2 mmol), (*S*)-**3e** (0.60 g, 2.4 mmol), Cy_2NH (0.90 mL, $d = 0.91\ g\,mL^{-1}$, 0.82 g, 4.5 mmol), and 1,4-dioxane (4 mL) was carried out for 6.5 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1 to 5:1) afforded (*S*)-**4e**[10] (0.36 g, 23% total yield (*R*)-**1e** to (*S*)-**4e**) as a solid; m.p. 87–89 °C (*n*-hexane/ethyl acetate); HPLC analysis (Chiralcel AD-H column, hexane/*i*PrOH = 90:10, 1.0 mL min^{-1}, $\lambda =$

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214 nm): $t_R = 15.29$ min (major) and $t_R = 13.16$ min (minor), 98% *ee*; $[\alpha]_D^{20} = +42.2$ ($c = 1.01$, $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): $\delta = 7.73$–7.67 (m, 2H, ArH), 7.53–7.37 (m, 3H, ArH), 7.34–7.20 (m, 5H, ArH), 6.15 (d, $J = 6.9$ Hz, 1H, CONH), 5.38–5.30 (m, 1H, C=CH), 5.03–4.91 (m, 1H, NCH), 4.91–4.83 (m, 2H, $C=CH_2$), 3.07 (dd, $J = 13.7$, 5.6 Hz, 1H, one proton in CH_2Ph), 2.98 ppm (dd, $J = 13.7$, 7.1 Hz, 1H, one proton in CH_2Ph); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 206.8$, 166.6, 137.1, 134.6, 131.4, 129.6, 128.5, 128.3, 126.8, 126.6, 92.0, 78.8, 48.3, 41.0 ppm; IR (neat): $\tilde{\nu} = 3305$, 2922, 1953, 1635, 1525, 1441, 1351, 1279, 1200, 1083, 1030 cm^{-1}; MS (EI): *m/z* (%): 263 (0.35) $[M]^+$, 105 (100).

Synthesis of 2-Phenyl-4,5-dihydrooxazoles 6

*4-Methyl-2-phenyl-5-(1-phenylvinyl)-4,5-dihydrooxazole (**6aa**)*

Typical Procedure II: $Pd(PPh_3)_4$ (11.9 mg, 0.01 mmol), **4a** (187.5 mg, 1.0 mmol), DMF (2 mL), **5a** (246.7 mg, 1.2 mmol), and DMF (2 mL) were added sequentially to a dry Schlenk tube that contained flame-dried K_2CO_3 (276.3 mg, 2.0 mmol) under argon. Then, the resulting solution was stirred in an oil bath that was preheated to 80 °C for 5 h. After the reaction was complete as monitored by TLC, the reaction mixture was diluted with diethyl ether (80 mL), and the resulting solution was washed with water (3×5 mL) and dried with anhydrous Na_2SO_4. After filtration and evaporation, the residue was purified by chromatography on a silica gel column (petroleum ether/ethyl acetate = 20:1) to afford *trans*-**6aa**[10] (235.7 mg, 89%) as an oil. ^1H NMR (300 MHz, $CDCl_3$): $\delta =$ 8.06–7.98 (m, 2H, ArH), 7.50–7.24 (m, 8H, ArH), 5.37 (s, 1H, one proton in $C=CH_2$), 5.35 (s, 1H, one proton in $C=CH_2$), 5.09 (d, $J =$ 6.6 Hz, 1H, OCH), 4.11–3.99 (m, 1H, NCH), 1.31 ppm (d, $J = 6.6$ Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 162.3$, 146.7, 138.1, 131.2, 128.4, 128.2, 128.1, 127.9, 127.7, 126.9, 113.0, 87.0, 67.9, 21.8 ppm; IR (neat): $\tilde{\nu} = 2969$, 1645, 1448, 1370, 1321, 1296, 1258, 1103, 1057, 1024 cm^{-1}; MS (EI): *m/z* (%): 263 (10.17) $[M]^+$, 131 (100).

The following compounds were prepared according to typical procedure II.

*2-Phenyl-5-(1-phenylvinyl)-4-propyl-4,5-dihydrooxazole (**6ba**)*

The reaction of K_2CO_3 (273.9 mg, 2.0 mmol), $Pd(PPh_3)_4$ (11.9 mg, 0.01 mmol), **4b** (215.4 mg, 1.0 mmol), DMF (2 mL), **5a** (245.9 mg, 1.2 mmol), and DMF (2 mL) was carried out for 6 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 20:1) afforded *trans*-**6ba** (256.4 mg, 88%) as an oil. ^1H NMR (300 MHz, $CDCl_3$): $\delta = 8.05$–7.99 (m, 2H, ArH), 7.52–7.26 (m, 8H, ArH), 5.36 (s, 1H, one proton in $C=CH_2$), 5.34 (s, 1H, one proton in $C=CH_2$), 5.13 (d, $J = 6.6$ Hz, 1H, OCH), 4.00 (q, $J = 6.4$ Hz, 1H, NCH), 1.70–1.48 (m, 2H, CH_2), 1.48–1.27 (m, 2H, CH_2), 0.85 ppm (t, $J = 7.2$ Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 162.4$, 147.3, 138.4, 131.3, 128.4, 128.3, 128.2, 127.9, 127.8, 127.2, 113.7, 85.6, 72.0, 38.3, 18.6, 13.9 ppm; IR (neat): $\tilde{\nu} = 2957$, 2930, 1651, 1494, 1449, 1324, 1290, 1061, 1025 cm^{-1}; MS (ESI): *m/z*: 292 $[M+H]^+$; HRMS: *m/z*: calcd for $C_{20}H_{21}NO$: 291.1623 $[M]^+$; found: 291.1624.

*4-Isopropyl-2-phenyl-5-(1-phenylvinyl)-4,5-dihydrooxazole (**6ca**)*

The reaction of K_2CO_3 (277.1 mg, 2.0 mmol), $Pd(PPh_3)_4$ (12.0 mg, 0.01 mmol), **4c** (215.6 mg, 1.0 mmol), DMF (2 mL), **5a** (245.3 mg, 1.2 mmol), and DMF (2 mL) was carried out for 10 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 20:1) afforded *trans*-**6ca**[10] (194.1 mg, 67%) as an oil. ^1H NMR (300 MHz, $CDCl_3$): $\delta = 8.06$–7.99 (m, 2H, ArH), 7.52–7.22 (m, 8H, ArH), 5.35 (s, 1H, one proton in $C=CH_2$), 5.31 (s, 1H, one proton in $C=CH_2$), 5.18 (d, $J = 6.3$ Hz, 1H, OCH), 3.87 (t, $J = 5.7$ Hz, 1H, NCH), 1.91–1.74 (m, 1H, CH), 0.88 ppm (dd, $J = 6.9$, 2.4 Hz, 6H, 2 CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 162.2$, 148.0, 138.4, 131.1, 128.3, 128.21, 128.15, 127.8, 127.7, 127.3, 114.1, 83.1, 77.7, 32.7, 18.3, 18.0 ppm; IR (neat): $\tilde{\nu} = 2958$, 1652, 1494, 1449, 1289, 1062, 1024 cm^{-1}; MS (EI): *m/z* (%): 291 (2.38) $[M]^+$, 144 (100).

*2,4-Diphenyl-5-(1-phenylvinyl)-4,5-dihydrooxazole (**6da**)*

The reaction of K_2CO_3 (278.7 mg, 2.0 mmol), $Pd(PPh_3)_4$ (11.6 mg, 0.01 mmol), **4d** (248.6 mg, 1.0 mmol), DMF (2 mL), **5a** (246.7 mg, 1.2 mmol), and DMF (2 mL) was carried out for 6.5 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 20:1) afforded *trans*-**6da** (280.5 mg, 86%) and then (petroleum ether/ethyl acetate = 5:1) *cis*-**6da** (20.2 mg, 6%). Data for *trans*-**6da**: Solid; m.p. 84–85 °C (*n*-hexane/ethyl acetate); ^1H NMR (300 MHz, $CDCl_3$): $\delta =$ 8.13 (d, $J = 7.5$ Hz, 2H, ArH), 7.60–7.43 (m, 3H, ArH), 7.39–7.21 (m, 8H, ArH), 7.10 (d, $J = 7.5$ Hz, 2H, ArH), 5.45 (s, 1H, one proton in $C=CH_2$), 5.43 (s, 1H, one proton in $C=CH_2$), 5.39 (d, $J = 6.9$ Hz, 1H, NCH), 5.03 ppm (d, $J = 6.6$ Hz, 1H, OCH); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta =$ 163.9, 146.6, 141.9, 137.7, 131.6, 128.6, 128.5, 128.4, 128.1, 127.6, 127.4, 127.2, 126.7, 113.6, 88.1, 75.9 ppm; IR (neat): $\tilde{\nu} = 1649$, 1492, 1450, 1322, 1287, 1251, 1066, 1021 cm^{-1}; MS (EI): *m/z* (%): 325 (1.44) $[M]^+$, 193 (100). Elemental analysis calcd (%) for $C_{23}H_{19}NO$: C 84.89, H 5.89, N 4.30; found: C 85.03, H 5.84, N 4.05. Data for *cis*-**6da**: Oil; ^1H NMR (300 MHz, $CDCl_3$): $\delta = 8.16$ (d, $J = 6.9$ Hz, 2H, ArH), 7.60–7.45 (m, 3H, ArH), 7.23–6.98 (m, 6H, ArH), 6.87 (d, $J = 7.5$ Hz, 2H, ArH), 6.78 (d, $J = 7.5$ Hz, 2H, ArH), 6.02 (d, $J = 9.9$ Hz, 1H, NCH), 5.51 (d, $J = 9.9$ Hz, 1H, OCH), 5.34 (s, 1H, one proton in $C=CH_2$), 5.20 ppm (s, 1H, one proton in $C=CH_2$); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 164.4$, 143.9, 138.9, 137.4, 131.7, 128.5, 128.4, 128.0, 127.6, 127.5, 127.3, 127.2, 126.1, 113.6, 83.8, 73.2 ppm; IR (neat): $\tilde{\nu} = 3052$, 2924, 1650, 1493, 1446, 1326, 1066, 1023 cm^{-1}; MS (EI): *m/z* (%): 325 (1.69) $[M]^+$, 193 (100); HRMS: *m/z*: calcd for $C_{23}H_{19}NO$: 325.1467 $[M]^+$; found: 325.1469.

*4-Benzyl-2-phenyl-5-(1-phenylvinyl)-4,5-dihydrooxazole (**6ea**)*

The reaction of K_2CO_3 (275.7 mg, 2.0 mmol), $Pd(PPh_3)_4$ (11.9 mg, 0.01 mmol), **4e** (265.0 mg, 1.0 mmol), DMF (2 mL), **5a** (244.6 mg, 1.2 mmol), and DMF (2 mL) was carried out for 5 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 20:1) afforded *trans*-**6ea**[10] (310.8 mg, 91%) as an oil; ^1H NMR (300 MHz, $CDCl_3$): $\delta = 8.02$ (d, $J = 6.9$ Hz, 2H, ArH), 7.53–7.37 (m, 3H, ArH), 7.26–7.04 (m, 10H, ArH), 5.25–5.17 (m, 3H, $C=CH_2$ and OCH), 4.30–4.20 (m, 1H, NCH), 3.09 (dd, $J = 13.8$, 5.7 Hz, 1H, one proton in CH_2Ph), 2.74 ppm (dd, $J = 13.5$, 8.1 Hz, 1H, one proton in CH_2Ph); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 163.1$, 146.8, 137.8, 137.3, 131.4, 129.5, 128.37, 128.36, 128.35, 128.25, 127.7, 126.9, 126.4, 113.5, 83.9, 73.2, 41.7 ppm; IR (neat): $\tilde{\nu} = 1648$, 1493, 1450, 1246, 1069, 1024 cm^{-1}; MS (EI): *m/z* (%): 339 (1.02) $[M]^+$, 248 (100).

*4-Cyclohexyl-2-phenyl-5-(1-phenylvinyl)-4,5-dihydrooxazole (**6fa**)*

The reaction of K_2CO_3 (275.8 mg, 2.0 mmol), $Pd(PPh_3)_4$ (11.8 mg, 0.01 mmol), **4f** (257.0 mg, 1.0 mmol), DMF (2 mL), **5a** (248.0 mg, 1.2 mmol), and DMF (2 mL) was carried out for 14 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 40:1 to 20:1) afforded *trans*-**6fa** (296.5 mg, 88%) as an oil. ^1H NMR (300 MHz, $CDCl_3$): $\delta = 8.05$–7.98 (m, 2H, ArH), 7.52–7.38 (m, 3H, ArH), 7.38–7.25 (m, 5H, ArH), 5.35 (s, 1H, one proton in $C=CH_2$), 5.31 (s, 1H, one proton in $C=CH_2$), 5.21 (d, $J = 6.3$ Hz, 1H, OCH), 3.85 (t, $J = 5.9$ Hz, 1H, NCH), 1.80–1.41 (m, 6H, Cy), 1.29–0.90 ppm (m, 5H, Cy); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 162.1$, 147.9, 138.5, 131.1, 128.26, 128.19, 128.15, 127.74, 127.65, 127.3, 114.3, 83.2, 76.7, 42.7, 28.73, 28.66, 26.3, 26.0 ppm; IR (neat): $\tilde{\nu} = 2922$, 2851, 1738, 1652, 1494, 1448, 1240, 1061, 1025 cm^{-1}; MS (EI): *m/z* (%): 331 (18.87) $[M]^+$, 84 (100); HRMS: *m/z*: calcd for $C_{23}H_{25}NO$: 331.1936 $[M]^+$; found: 331.1939.

*5-[1-(4-Acetophenyl)vinyl]-4-cyclohexyl-2-phenyl-4,5-dihydrooxazole (**6fb**)*

The reaction of K_2CO_3 (274.9 mg, 2.0 mmol), $Pd(PPh_3)_4$ (11.7 mg, 0.01 mmol), **4f** (255.4 mg, 1.0 mmol), DMF (2 mL), **5b** (294.5 mg, 1.2 mmol), and DMF (2 mL) was carried out for 15 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 10:1) afforded *trans*-**6fb** (317.3 mg, 85%) as a solid; m.p. 108–109 °C (*n*-hexane/ethyl acetate); ^1H NMR (300 MHz, $CDCl_3$): $\delta = 8.03$–7.97 (m, 2H, ArH), 7.91 (d, $J = 8.4$ Hz, 2H, ArH), 7.53–7.38 (m, 5H, ArH), 5.45 (s, 1H, one proton in $C=CH_2$), 5.40 (s, 1H, one proton in $C=CH_2$), 5.23

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(d, J = 6.3 Hz, 1H, OCH), 3.83 (t, J = 6.0 Hz, 1H, NCH), 2.58 (s, 3H, $COCH_3$), 1.82–1.43 (m, 6H, Cy), 1.30–0.90 ppm (m, 5H, Cy); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 197.6, 162.1, 147.2, 143.4, 136.4, 131.3, 128.4, 128.3, 128.2, 127.61, 127.56, 116.3, 83.1, 76.9, 42.8, 28.9, 28.8, 26.6, 26.3, 26.0 ppm; IR (neat): $\tilde{\nu}$ = 2921, 2852, 1679, 1646, 1603, 1451, 1354, 1326, 1264, 1067, 1006 cm^{-1}; MS (EI): m/z (%): 373 (28.61) $[M]^+$, 290 (100). Elemental analysis calcd (%) for $C_{25}H_{27}NO_2$: C 80.40, H 7.29, N 3.75; found: C 80.32, H 7.16, N 3.56.

*4-Cyclohexyl-2-phenyl-5-(1-m-tolylvinyl)-4,5-dihydrooxazole (**6fc**)*

The reaction of K_2CO_3 (276.0 mg, 2.0 mmol), $Pd(PPh_3)_4$ (11.9 mg, 0.01 mmol), **4f** (255.1 mg, 1.0 mmol), DMF (2 mL), **5c** (263.0 mg, 1.2 mmol), and DMF (2 mL) was carried out for 24 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 50:1) afforded *trans*-**6fc** (296.3 mg, 86%) as an oil. ^{1}H NMR (300 MHz, $CDCl_3$): δ = 8.05–7.98 (m, 2H, ArH), 7.49–7.34 (m, 3H, ArH), 7.22–7.04 (m, 4H, ArH), 5.32 (s, 1H, one proton in C=CH_2), 5.28 (s, 1H, one proton in C=CH_2), 5.21 (d, J = 6.0 Hz, 1H, OCH), 3.85 (t, J = 5.9 Hz, 1H, NCH), 2.29 (s, 3H, CH_3), 1.81–1.40 (m, 6H, Cy), 1.30–0.79 ppm (m, 5H, Cy); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 162.1, 148.0, 138.4, 137.8, 131.1, 128.5, 128.2, 128.1, 127.8, 124.4, 114.0, 83.3, 76.8, 42.8, 28.8, 28.7, 26.3, 26.0, 21.29, 21.28 ppm; IR (neat): $\tilde{\nu}$ = 2922, 2851, 1652, 1581, 1494, 1449, 1323, 1288, 1062, 1025 cm^{-1}; MS (EI): m/z (%): 345 (25.32) $[M]^+$, 262 (100); HRMS: m/z: calcd for $C_{24}H_{27}NO$: 345.2093 $[M]^+$; found: 345.2092.

*5-[1-(4-Bromophenyl)vinyl]-4-cyclohexyl-2-phenyl-4,5-dihydrooxazole (**6fd**)*

The reaction of K_2CO_3 (276.9 mg, 2.0 mmol), $Pd(PPh_3)_4$ (11.9 mg, 0.01 mmol), **4f** (255.8 mg, 1.0 mmol), DMF (2 mL), **5d** (339.8 mg, 1.2 mmol), and DMF (2 mL) was carried out for 9 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 30:1) afforded *trans*-**6fd** (337.0 mg, 82%) as an oil. ^{1}H NMR (300 MHz, $CDCl_3$): δ = 8.03–7.96 (m, 2H, ArH), 7.51–7.36 (m, 5H, ArH), 7.25–7.17 (m, 2H, ArH), 5.36 (s, 1H, one proton in C=CH_2), 5.29 (s, 1H, one proton in C=CH_2), 5.16 (d, J = 6.3 Hz, 1H, OCH), 3.82 (t, J = 6.0 Hz, 1H, NCH), 1.82–1.40 (m, 6H, Cy), 1.30–0.84 ppm (m, 5H, Cy); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 162.0, 146.9, 137.4, 131.5, 131.2, 129.0, 128.3, 128.2, 127.6, 122.0, 115.3, 83.2, 76.82, 76.77, 42.8, 28.9, 28.8, 26.3, 26.0 ppm; IR (neat): $\tilde{\nu}$ = 2922, 2851, 1739, 1652, 1487, 1449, 1323, 1288, 1061, 1024, 1009 cm^{-1}; MS (EI): m/z (%): 411 (7.12) $[M(^{81}Br)]^+$, 409 (6.86) $[M(^{79}Br)]^+$, 105 (100); HRMS: m/z: calcd for $C_{23}H_{24}{}^{79}BrNO$: 409.1041 $[M]^+$; found: 409.1044.

*4-Cyclohexyl-2-phenyl-5-[1-(thiophen-2-yl)vinyl]-4,5-dihydrooxazole (**6fe**)*

The reaction of K_2CO_3 (276.4 mg, 2.0 mmol), $Pd(PPh_3)_4$ (11.8 mg, 0.01 mmol), **4f** (256.0 mg, 1.0 mmol), DMF (2 mL), **5e** (249.2 mg, 1.2 mmol), and DMF (2 mL) was carried out for 12 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 40:1) afforded *trans*-**6fe** (290.8 mg, 85%) as an oil. ^{1}H NMR (300 MHz, $CDCl_3$): δ = 8.05–7.99 (m, 2H, ArH), 7.53–7.37 (m, 3H, ArH), 7.20 (dd, J = 5.1, 0.9 Hz, 1H, thienyl), 7.05 (dd, J = 3.5, 1.1 Hz, 1H, thienyl), 6.96 (dd, J = 5.1, 3.6 Hz, 1H, thienyl), 5.52 (s, 1H, one proton in C=CH_2), 5.29 (s, 1H, one proton in C=CH_2), 5.14 (d, J = 6.9 Hz, 1H, OCH), 3.99 (t, J = 6.2 Hz, 1H, NCH), 1.91–1.51 (m, 6H, Cy), 1.34–0.98 ppm (m, 5H, Cy); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 162.3, 140.9, 140.3, 131.3, 128.3, 127.7, 127.4, 125.2, 124.6, 114.3, 84.1, 76.8, 43.0, 29.2, 28.6, 26.4, 26.15, 26.09 ppm; IR (neat): $\tilde{\nu}$ = 2922, 2851, 1652, 1448, 1324, 1228, 1060, 1024 cm^{-1}; MS (EI): m/z (%): 337 (21.56) $[M]^+$, 105 (100); HRMS: m/z: calcd for $C_{21}H_{23}NOS$: 337.1500 $[M]^+$; found: 337.1497.

*(4S,5S)-4-Cyclohexyl-2-phenyl-5-(1-phenylvinyl)-4,5-dihydrooxazole ((S,S)-**6fa**)*

The reaction of K_2CO_3 (138.0 mg, 1.0 mmol), $Pd(PPh_3)_4$ (6.0 mg, 0.005 mmol), (*S*)-**4f** (127.0 mg, 0.5 mmol), **5a** (123.4 mg, 0.6 mmol), and DMF (2 mL) was carried out for 11.5 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 40:1 to 20:1) afforded (*S,S*)-**6fa** (140.1 mg, 85%) as an oil. HPLC analysis (Chiralcel AD-H column, hexane/*i*PrOH = 95:5, 1.0 mL min^{-1}, λ = 214 nm): t_R = 6.03 min (major) and t_R = 7.58 min (minor), 99% *ee*; $[\alpha]_D^{23}$ = −21.2 (c = 1.00, $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 8.04–7.98 (m, 2H, ArH), 7.52–7.38 (m, 3H, ArH), 7.38–7.24 (m, 5H, ArH), 5.35 (s, 1H, one proton in C=CH_2), 5.31 (s, 1H, one proton in C=CH_2), 5.22 (d, J = 6.0 Hz, 1H, OCH), 3.85 (t, J = 5.9 Hz, 1H, NCH), 1.81–1.40 (m, 6H, Cy), 1.31–0.85 ppm (m, 5H, Cy); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 162.2, 148.0, 138.6, 131.2, 128.4, 128.3, 128.2, 127.84, 127.80, 127.4, 114.4, 83.3, 76.9, 42.8, 28.84, 28.78, 26.4, 26.1 ppm; IR (neat): $\tilde{\nu}$ = 2922, 2851, 1652, 1494, 1448, 1323, 1289, 1062, 1025 cm^{-1}; MS (EI): m/z (%): 331 (19.68) $[M]^+$, 248 (100); HRMS: m/z: calcd for $C_{23}H_{25}NO$: 331.1936 $[M]^+$; found: 331.1932.

*(4S,5S)-2-Phenyl-5-(1-phenylvinyl)-4-propyl-4,5-dihydrooxazole ((S,S)-**6ba**)*

The reaction of K_2CO_3 (138.1 mg, 1.0 mmol), $Pd(PPh_3)_4$ (6.1 mg, 0.005 mmol), (*S*)-**4b** (106.7 mg, 0.5 mmol), **5a** (123.2 mg, 0.6 mmol), and DMF (2 mL) was carried out for 6 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 20:1) afforded (*S,S*)-**6ba** (125.4 mg, 83%, 96% purity) as an oil. HPLC analysis (Chiralcel AD-H column, hexane/*i*PrOH = 95:5, 1.0 mL min^{-1}, λ = 214 nm): t_R = 5.86 min (major) and t_R = 7.21 min (minor), 95% *ee*; $[\alpha]_D^{19}$ = +14.5 (c = 1.02, $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 8.05–7.97 (m, 2H, ArH), 7.53–7.25 (m, 8H, ArH), 5.36 (s, 1H, one proton in C=CH_2), 5.34 (s, 1H, one proton in C=CH_2), 5.13 (d, J = 6.6 Hz, 1H, OCH), 4.00 (q, J = 6.4 Hz, 1H, NCH), 1.72–1.48 (m, 2H, CH_2), 1.48–1.27 (m, 2H, CH_2), 0.85 ppm (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 162.4, 147.4, 138.4, 131.3, 128.5, 128.3, 128.2, 128.0, 127.8, 127.2, 113.7, 85.6, 72.0, 38.3, 18.7, 14.0 ppm; IR (neat): $\tilde{\nu}$ = 2958, 2930, 1646, 1520, 1489, 1449, 1267, 1062, 1025 cm^{-1}; MS (ESI): m/z: 292 $[M+H]^+$; HRMS: m/z: calcd for $C_{20}H_{21}NO$: 291.1623 $[M]^+$; found: 291.1627.

*(4S,5S)-4-Benzyl-2-phenyl-5-(1-phenylvinyl)-4,5-dihydrooxazole ((S,S)-**6ea**)*

The reaction of K_2CO_3 (138.1 mg, 1.0 mmol), $Pd(PPh_3)_4$ (6.2 mg, 0.005 mmol), (*S*)-**4e** (131.6 mg, 0.5 mmol), **5a** (122.1 mg, 0.6 mmol), and DMF (2 mL) was carried out for 8.5 h. Purification by chromatography on a silica gel column (petroleum ether/ethyl acetate = 20:1) afforded (*S,S*)-**6ea**[10] (134.9 mg, 80%) as an oil. HPLC analysis (Chiralcel AD-H column, hexane/*i*PrOH = 90:10, 1.0 mL min^{-1}, λ = 214 nm): t_R = 6.13 min (major) and t_R = 8.59 min (minor), 98% *ee*; $[\alpha]_D^{29}$ = +61.3 (c = 0.97, $CHCl_3$), ref. [10] $[\alpha]_D^{25}$ = +22.93 (c = 0.41, $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 8.02 (d, J = 6.9 Hz, 2H, ArH), 7.54–7.37 (m, 3H, ArH), 7.26–7.04 (m, 10H, ArH), 5.25–5.16 (m, 3H, C=CH_2 and OCH), 4.30–4.21 (m, 1H, NCH), 3.09 (dd, J = 13.5, 5.4 Hz, 1H, one proton in CH_2Ph), 2.74 ppm (dd, J = 13.6, 7.9 Hz, 1H, one proton in CH_2Ph); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 163.1, 146.8, 137.9, 137.3, 131.4, 129.6, 128.39, 128.37, 128.36, 128.27, 127.7, 126.9, 126.4, 113.5, 84.0, 73.2, 41.8 ppm; IR (neat): $\tilde{\nu}$ = 1649, 1493, 1450, 1322, 1287, 1241, 1056, 1023 cm^{-1}; MS (EI): m/z (%): 339 (0.90) $[M]^+$, 248 (100).

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[11] The crystal data of compound **4da**: $C_{23}H_{19}NO$; MW = 325.39; monoclinic; space group $P21/c$; Mo Kα; final R indices $[I > 2\sigma(I)]$; $R_1 = 0.0740$; $wR_2 = 0.2143$; $a = 12.1014(19)$ Å, $b = 14.615(2)$ Å, $c = 10.1483(16)$ Å; $\alpha = 90.00°$, $\beta = 100.454(5)°$, $\gamma = 90.00°$; $V = 1765.0(5)$ Å^3; $T = 173(2)$ K; $Z = 4$; reflections collected/unique: 19765/2011 ($R_{int} = 0.0927$); number of observations $[> 2\sigma(I)]$ 3101, parameters 226. CCDC982501 (**4da**) contains the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via http://www.ccdc.cam.ac.uk/data_request/cif.

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